

Sensorized Phantom For Characterizing Large Area Deformation of Soft Bodies for Medical Applications

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Abstract—Robotics is a key technology to improve and support the medical services by increasing efficiency and improving quality of care. One particular challenge is medical palpation, where a physicians uses their hands to manually feel and identify abnormalities or tumors. This is a challenging procedure that takes many years to learn. A soft robotic phantom which could provide feedback on a trainees palpation technique could enable medical professionals to learn more easily. Furthermore, this would allow quantitative information about how practitioners perform the technique to be gained. This can be used to inform and move towards robotic palpation that mirrors the human performance. In this work, we demonstrate an approach for developing a large-area sensorized phantom that uses sensor morphology to sense over a large areas. A reconstruction algorithm allows the depth of palpation to be measured and the location to be identified with a precision of $\pm 5mm$. This phantom was used to measure deformation from human palpation, with the resultant information used to replicate the behaviour with a robot hand. To the authors knowledge this is the first sensorized phantom that characterizes large scale deformation from human palpation to inform the performance of the procedure by a robot.

I. INTRODUCTION

Medical Robotics is a rapidly evolving field, partially due to the advent of soft robotics [1]. There are two main areas where medical robotics can offer significant advances. Firstly, to enhance or extend the capabilities of human medical procedures with the development of training robot simulators or ‘phantoms.’ And secondly to develop robots which can perform medical examinations and procedures [2]. The development of robot training devices enables safe and efficient training of future medical professionals and can also be used for developing and testing robots to perform medical procedures [3]. Methods such as haptic feedback and virtual reality simulations are already becoming ubiquitous tools, in particular for laproscopic surgery [3]. However, there is limited development of medical phantoms which can provide analysis of tactile based examinations.

Medical palpation is one area where there is a significant need for a training aids and also for robot doctors, to both allow for quicker training and reduce waiting times for such examinations [4]. Defined as the process of using ones hands to check the body to help perceive or diagnose a disease or illness, it is a key part physical examination required

for many diagnoses [5]. Haptic sense is indispensable for the detection of subsurface tumors, which makes palpation a difficult skill to demonstrate and learn. Additionally, the methods by which medical professionals perform palpation are poorly understood, which makes developing robotic palpation systems challenging. Having a simulator that can record and show kinematics, touch, and force feedback allows trainees to learn the methodology of a procedure, whilst also recording and identifying how palpation is performed.

To address this limitation, we have developed a large area sensorized soft phantom which allows the specific characteristic of palpation to be reconstructed. The development of large area sensorized bodies is challenging as the continuous soft surface can deform in a potentially infinite manner. The sensors must not restrict the movement of the soft body whilst also capturing this continuously varying deformation; a typical one-to-one mapping of degree-of-freedom and sensor is not an approach which can be used in this situation. Methods for developing sensorized bodies which can use a finite number of sensors are required.

To overcome these challenges, we propose using a new sensing technology, Conductive Thermoplastic Elastomer (CTPE), and developing scaleable approaches for the sensor morphology to allow large area coverage. This approach is scale-able and does not restrict the intrinsic flexibility of the soft body. Secondly, we introduce a model of the deformation caused by palpation and propose an algorithm for reconstructing the deformation.

To demonstrate these approaches we show the ability to reconstruct human palpation characteristics (location, depth and area). We show how the information gained can be used to developing a control strategy for a compliant robot hand to perform palpation with a similar profile to that of humans. Although, this is a demonstration system for a non-specific part of the body it allows characterization and demonstration of the key principles.

In this paper the following sections provides background on medical palpation and discusses current state of the art technologies used to assist with palpation examinations. Following this, the construction of the phantom and the sensor reconstruction algorithm is presented. The experimental results and analysis characterizing the phantom are then given. Finally, using experimental results from human palpation, a robotic hand is used to replicate the deformation caused by humans. The concluding discussion examines the effectiveness of the system and proposes for future work.

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II. BACKGROUND

A. Medical Palpation

Palpation is a diagnostic technique which uses the interaction between the medical professional's hands and the patient's body to aid detection of abnormalities [5]. The technique can be used for identifying tumours or other abnormalities in soft tissues such as the abdomen or breast [6], [7]. The physical interaction between doctors hands and the patient's body, and the methods with which they identify abnormalities is poorly understood.

Broadly, palpation can be split into light and deep [8]. Light palpation is used to feel for surface abnormalities [9]. The skin is depressed by 1-2cm with the finger pads, using the lightest touch possible and typically with three fingers. This is used to assess for skin properties including texture, tenderness, temperature, masses and pulsations. Deep palpation is used to feel internal organs and masses for size, shape, tenderness, symmetry, and mobility. The skin can be depressed by up to 4-5cm with firm, deep pressure. Typically, one hand is used on top of the other to exert firm pressure using a three finger hand position. For these two main types of palpation, the key deformation characteristics which describe palpation include: location, depth, width, frequency and rate at which the deformation occurs.

B. Large Area Soft Sensing

A sensorized phantom need to be deformable and thus, sensing elements must be included over a large area without compromising flexibility or the inherent material properties. There are number of approaches to large area soft sensing. Soft iconic liquid based sensors have been developed that allow large area sensing of deformation [10], [11]. Large area flexible force sensors have also been developed; however, these do not always provide deformation information and require a rigid component to achieve a reaction force [12]. Although the development of methods for detecting deformation of larger soft surfaces has been increasing, these often require the inclusion of large rigid objects, making them unsuitable for this application [13], [14].

III. METHODS

A. Sensor Material

To develop the large area soft sensorized phantom, a piezoresistive material, Conductive Thermoplastic Elastomer (CTPE) has been used. CTPE is a thermoplastic elastic matrix which is homogeneously mixed with carbon black powder under high pressure and temperature [15]. This process produces an electrically conductive material, which has been shown to have a linear response to strain [15]–[19].

CTPE can be extruded into a wide range of shapes or morphologies, including highly elastic fibres typically of 0.3mm diameter, which can undergo strains of above 80% without reaching their tensile limit. The flexibility of CTPE is a key advantage as it allows the specific morphology and integration into a soft body to be varied. The CTPE sensor morphology can be designed to detect a particular stimuli

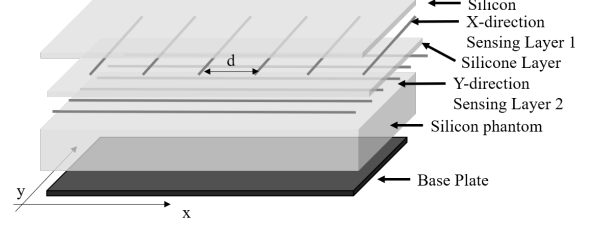


Fig. 1. Structure of the phantom showing the silicone layers and the sensor layer and the line sensor morphology used.

with a given accuracy or precision. For an starting length of CTPE x_0 , the relative change in electrical resistance per unit of mechanical strain can be described by:

$$\frac{\Delta R}{R_0} = \sigma \epsilon \quad (1)$$

where R_0 is the base resistance of the sensor, which undergoes a strain of $\epsilon = x/x_0$ where the CTPE has an experimentally determined gauge factor of σ . This relationship has been previously verified [16].

B. Phantom Sensorization

The sample phantom has been developed by casting layers of EcoFlex 00-10 sandwiched together with sensing layers of CTPE. The height of the bottom silicone layer determines the range of possible deformation and compression of the phantom. This is followed by a sensing layer. The sensing layer consists of a x-direction sensing layer, a thin layer of silicone and then the y-direction sensing layer, sandwiched together using Dow-Corning Silicone adhesive (Fig. 1). Each sensor layer consists of an array of CTPE sensors, each straight lines spanning the length of the phantom, separated by the distances d_x and d_y in the two respective directions. Each layer enables the respective localization of the deformation in the x and y direction. For n sensors in each of the x and y directions, each with a sensor length, l , the sensor morphology can be described by:

$$\mathbf{M}_{nx} = [nd, l], \mathbf{M}_{ny} = [l, nd] \quad (2)$$

The change in resistance in each sensor reflects the change in length caused by the deformation. Due to the soft nature of the phantom, localized point interaction causes larger area deformation on the surface thus, a discrete set of linear sensors can be used to measure the deformation of a continuous surface. The separation, d , between the sensor lines must be chosen for the specific application to ensure that the minimum size of the deformation, d_{min} , is greater than the separation between the sensors ($d_{min} > d$). The sensor separation should be optimized to reduce the number of sensors whilst maximizing the sensitivity. This expands upon previous work on sensor morphology [17]–[19].

Each CTPE sensor is connected via terminal block to a wired connection, with each sensor forming a potential divider with a matched resistor to provide an analog output. The analogue output from each sensor and resistor pair is

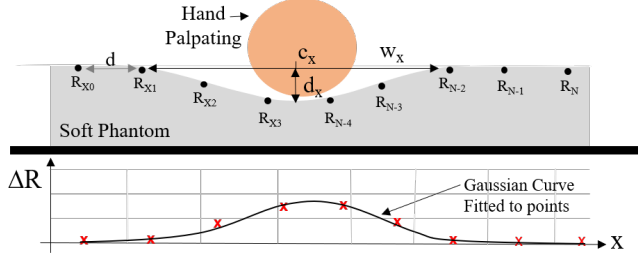


Fig. 2. 2D cross section of the phantom showing the deformation and the sensor morphology.

connected to a 10-bit I2C ADC, which is then connected to a micro controller to allow real time processing.

C. Deformation Modelling & Reconstruction

Palpation is assumed to cause a single point deformation of the soft body which can be approximated by a 3D deformation profile of an inverted bivariate Gaussian. Thus, in both the x and y directions, the deformation can be approximated as a 2D Gaussian. The parameters of this 2D Gaussian can be identified from the discrete deformation information from each sensor, as shown in Fig.2. For each sensor, the relative change in resistance from the undeformed base resistance is used such that a vector of points represents the change in length across a 2D slice:

$$R_x = [nd, \Delta R] \quad (3)$$

The two 2D Gaussian have the form:

$$D_x = a_x e^{-\frac{(x-b_x)^2}{2c_x^2}} \quad (4)$$

$$D_y = a_y e^{-\frac{(y-b_y)^2}{2c_y^2}} \quad (5)$$

A Gaussian has been chosen as empirically it fits the deformation experienced during palpation, and allows the depth, location and area characteristics of palpation to be determined. Using the constants from the fitted Gaussian, it is possible to firstly identify the centre of the Gaussian:

$$(m_x, m_y) = (b_x, b_y) \quad (6)$$

and to approximate the width of the deformation around the central point:

$$(w_x, w_y) = (2c_x, 2c_y) \quad (7)$$

To determine the depth of palpation, the central co-ordinate is used to evaluate the maximum change in resistance in the corresponding direction at the central co-ordinates of the deformation:

$$\Delta R_{X_c} = D_x(b_x = m_x, c_x, a_x) \quad (8)$$

$$\Delta R_{Y_c} = D_y(b_y = m_y, c_y, a_y) \quad (9)$$

From which the change in length can then be determined:

$$\Delta L_{X_c} = \frac{\Delta R_{X_c} x_0}{R_0}, \quad \Delta L_{Y_c} = \frac{\Delta R_{Y_c} x_0}{R_0} \quad (10)$$

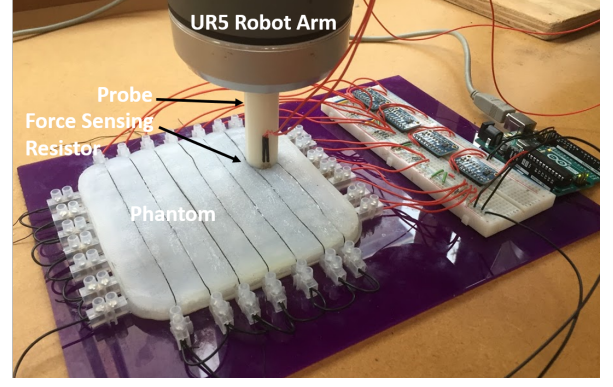


Fig. 3. Experimental setup showing a probe attached to a robot arm to allow characterization of the sensing abilities of the phantom.

Thus, to find the depth, the line integral over the function can be used to solve for the value of d:

$$\int_0^L D_x ds - L_0 \quad \text{where } ds = \sqrt{\frac{dx^2}{dl} + \frac{dy^2}{dl}} dl \quad (11)$$

with a corresponding solution for the y-sensing direction. The line integral is performed using numerical integration methods in Matlab, which allows the depth to be determined. The depth is then found by averaging the two depths found for the two Gaussians.

Although this choice of modelling the palpation with an inverted bivariate Gaussian is somewhat arbitrary, it captures the key spatial parameters of palpation: depth and width of the palpation area.

IV. EXPERIMENTAL SETUP

To characterize the sensorized phantom and the reconstruction algorithm, a UR5 robot arm with a probe attached has been used. This allows repeatable and numerous palpation style movements to be performed to characterize the sensorized phantom. The experimental setup is shown in Fig. 3, a force sensing resistor (FSR) has been fitted to the end of the probe to provide a ground truth of the force applied to the phantom. By knowing the area of contact, a ground truth for the depth of palpation can then be obtained from the results from the FSR. In the second set of experiments, humans perform palpation on the phantom. In the final case-study, a robot hand attached to the UR5 arm was used to mimic human palpation, with the arm providing 'wrist' movement to a compliant 3D printed hand.

V. EXPERIMENTAL RESULTS

In this section, the performance of the sensorized phantom is characterized to capture the abilities of the sensing system and the reconstruction algorithm.

A. Localization Characterization

To test the abilities of the phantom to determine the central location of the deformation caused by palpation, the probe was moved in a grid pattern with 5mm separation between

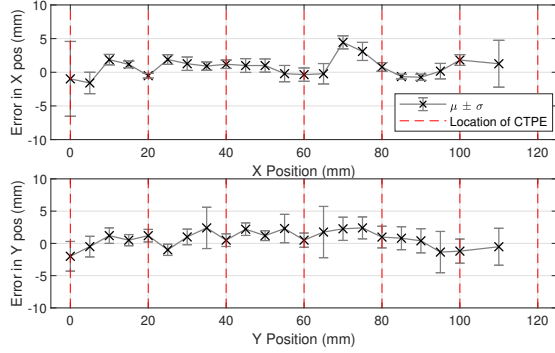


Fig. 4. Error in localization of palpation, in both the x and y direction, where a (x-y) grid with 5mm separation was palpated across the phantom. Each point was tested five times and the average and standard deviation shown.

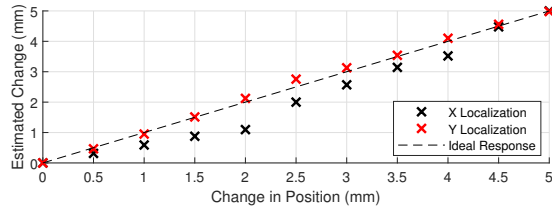


Fig. 5. Over the surface of the phantom the change in position from an initial starting was recorded, for steps of 0.5mm. This was repeated five times from different starting points.

points in both the x and y directions over the entire surface of the phantom, palpating with a fixed depth and rate (8mm and 0.1ms^{-1}) at each location five times. Using (6), the central location of each palpation was found. The results for the x and y direction are shown in Fig. 5. This shows how the localization accuracy is the highest closest to the sensor strings, although despite this the centre of location can be determined with an accuracy of within $\pm 5\text{mm}$. The error bars show that the precision, typically within $\pm 2\text{mm}$, is slightly less for the secondary sensing layer (the Y grid). Thus the accuracy of the Y-direction grid is on average lower than that of the x-direction. This results in the observed under-estimations of depth of palpation as the deformation experienced is slightly lower than the top layer.

Typically, the nodules or abnormalities which must be detected are 0.8-1 cm in diameter and thus, this resolution is sufficient to allow detection of the palpation location. Reducing the sensor separation could further increase the precision.

B. Localization Resolution

In addition to characterizing the localization over the entire body, the minimum distance between points which could be uniquely identified, the localization resolution, was investigated. The results are shown in Fig. 5 and show that the y-direction layer had a marginally higher resolution. This is thought to be due to the fact that the y sensing layer is on top of the x layer.

The results show that it is possible to detect changes in position which are of an order of 1mm providing an

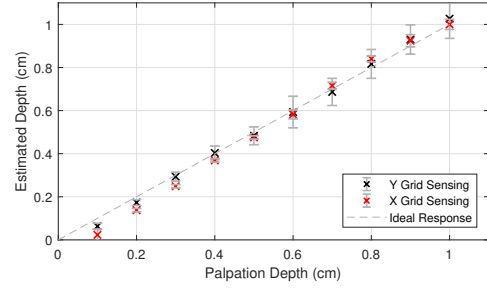


Fig. 6. Estimated depth and the true depth as determined by the robotic phantom. Was tested at five different points, with the test repeated four times at each point.

high resolution. By reducing the separation, d , between the sensors, the resolution of the phantom can be significantly increased at the expense of a greater number of sensors.

C. Depth Estimation

To characterize the ability of the phantom to detect the depth and width, the probe was moved to five different locations over the surface of the phantom. At each of these locations the probe was lowered by varying amounts between 1mm and 1cm, with a 1 second pause before being raised. The experiment was repeated four times at each location. The FSR on the probe was used to measure the force. The FSR was calibrated using a set of measuring scales. By using the known area of the probe, the ground truth for depth can then be obtained from this force measurement. The estimated depth from (11) and the ground truth are given in (Fig. 6).

D. Temporal Characteristics

Another key characterization experiment is determining the frequency response of the system. This limits the maximum sensory input which can be accurately measured. For palpation, maximum frequencies are around 1.5Hz (determined from experiments with practitioners), however, for other applications the frequency response may be more critical. Fig. 7 shows the effects that frequency has on the magnitude of the response. This was determined by palpating the phantom using the probe mounted on the end of the UR5 arm which is moved at a fixed speed and depth to replicate palpation. At each frequency, the response from both the phantom and the force sensing resistor is recorded 10 times. The FSR on the probe is used to provide the ground truth, whilst the frequency response and magnitude is then found from the sensorized phantom. The results demonstrate that there is minimum attenuation of the palpation at the frequencies of interest (around 1 Hz). The damping of the soft body does limit the range of high frequencies, however, for this application the response is appropriate and will allow detection of palpation. In addition to the effect that frequency has on the magnitude, the effect sensitivity of the system to the rate of changes, notionally the phase response, was investigated (Fig. 7).

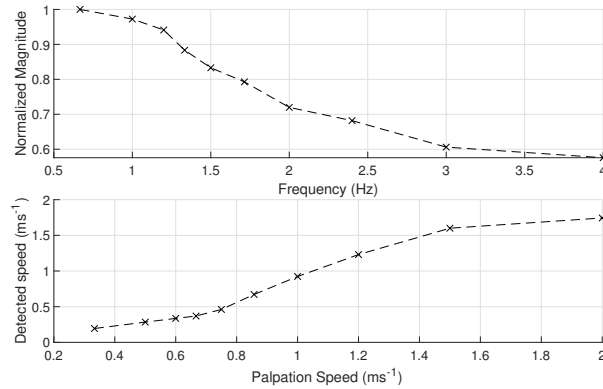


Fig. 7. The effect of increasing frequency on the magnitude of the sensing response.

TABLE I
CHARACTERISTICS OF HUMAN PALPATION OBTAINED FROM THE
SENSORIZED PHANTOM.

Palpation Characteristic	Light Palpation	Deep Palpation
Depth	3 mm $\pm$ 0.7mm	6 mm $\pm$ 1.2mm
X Direction Width	64 mm $\pm$ 12mm	67 mm $\pm$ 10mm
Y Direction Width	29 mm $\pm$ 4mm	32 mm $\pm$ 5mm
Frequency of palpation	0.8Hz $\pm$ 0.15	0.3Hz $\pm$ 0.1

E. Human Palpation Experiments

After characterization, the next experiments seek to understand human palpation and to demonstrate the capabilities of the phantom. These experiments provide only an initial demonstration, future work with expert practitioners will investigate the performance further. We consider both single handed ‘light’ palpation and two handed deeper palpation performed using a standard three finger method. The person performing the palpation motion, was provided with instruction of how to perform palpation and asked to explore the phantom using palpation, with the goal of exploring the phantom for inflammation. This replicates the scenario of exploring a patient’s abdomen for inflammation. Fig. 8 shows images of the light palpation, the ground truth and estimated location and depth. The hand was moved to a measured position to allow the true position to be determined, and the depth determined by measuring the force applied to the phantom using a set of accurate scale. The ground truth of the location was found by using a camera mounted above the phantom. Table I shows the characteristics of light and deep palpation averaged across 40 different palpation experiments for both light (one handed) and deep (two handed) palpation performed using medical palpation methods. These experiments were performed by two non-experts (students) each performing 50 experiments to demonstrate the system. Interestingly the results showed how the palpation depth of human palpation shows reasonable consistency.

VI. CASE STUDY: ROBOTIC PALPATION

Using the results from human palpation, we now aim to replicate this palpation using a robot hand. For this we use a 3D printed anthropomorphic hand including only the

bones and ligaments. This has been printed on a multi-material Stratsys 3D printer. The ligaments were printed with a lower-stiffness material such that the joints have some flexibility [20]. The ligament stiffness has been set to reflect that of a human hand held rigid, and then, as with human palpation the motion is achieved through wrist actuation, using the passive static behaviours of the hand. The hand is mounted on the end of a UR5 robot arm (Fig. 9). The hand shows some passive stiffness, however, is controlled only by the wrist movements.

The wrist movement can be described by, $w = [\Delta z, \dot{z}, f]$, reflecting the depth of palpation (Δz), rate ($\dot{z}$) and frequency (f). The parameters were iteratively improved using proportional feedback until they best matched that of human palpation for both light and deep palpation.

To demonstrate how the phantom can be used to allow human examinations to inform robot examinations, the deformation characteristics for human palpation and the tuned robot palpation is shown in Fig. 10. These results show that a similar response can be obtained with the passive stiffness of the hand contributing to the palpation behaviour, allowing a profile close to that of human to be achieved with minimal control. By integrating force sensing into the hand, it would be possible to exploit this passive stiffness to detect small changes in material properties which arise from the presence of abnormalities.

VII. DISCUSSION & CONCLUSION

In this work a method of integrating CTPe into a phantom has been proposed to create a sensorized phantom. By implementing the proposed grid morphology, and using the reconstruction method, it has been shown that it is possible to localize the palpation with an accuracy of $\pm 5mm$. Additionally, it is also possible to provide an indication of the width, depth and rate of palpation. Further work investigating the identification of palpation characteristics of a wider group of trained experts should be undertaken. The ability to use the phantom to provide specific training should also be explored, by testing the abilities of the system to aid and benefit novice practitioners.

The integration of the sensorization method into a full body phantom would provide the ability to achieve increasing human and robot palpation. This would allow full understanding of how humans interact with a patient to both help assist with the development of robotic medical examinations, making experimentation safer and more efficient to develop.

The replication of palpation robots shows the benefits of combining a phantom with a robotic manipulator, to close the loop and allow feedback to be used to optimize the robots behaviour. Although in the example proposed it has been shown how replication can be achieved, expanding this to provide methods of improving robot and human palpation would further extend the capabilities of the system.

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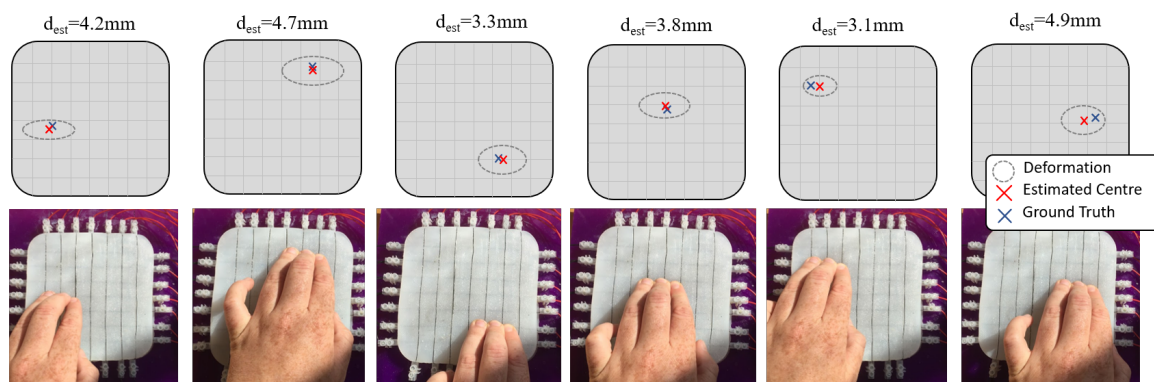


Fig. 8. Demonstration of the phantom showing the estimated location and depth in comparison to pictures and the ground truth.

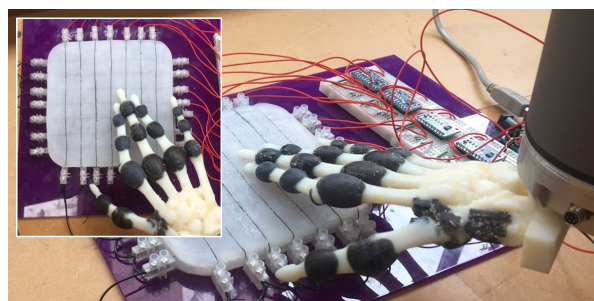


Fig. 9. Passive robot hand undergoing palpation experiments, showing the attachment of the hand to the robot.

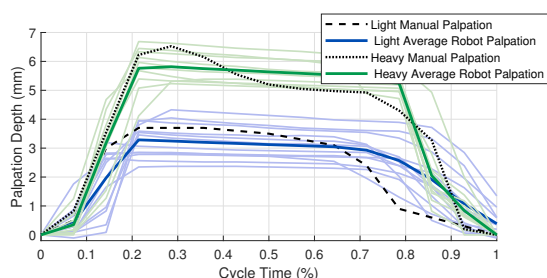


Fig. 10. Comparison between the depth of palpation over the palpation cycle for the robot hand and human hand for both light and deep palpation.

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